Name:

# Final Exam Solutions

**Instructions.** The only tools you are permitted to use are pencils, pens, erasers, and your mind. No electronic devices! You have 2.5 hours. When you finish, please hand your exam to Sunny in his office (Tutt Science 234). Good luck!

**Problem 1** (True-false, 10 points)**.** You get 1 point for each correct answer. No explanations required.

(1) Any set of 4 vectors in $\mathcal{P}_2$ must be linearly dependent. **T** F

(2) $\mathbb{R}^4$ and $\mathcal{M}_{2\times 2}$ are isomorphic vector spaces. **T** F

(3) If $B = \langle 1, 1+x, 1+x+x^2 \rangle$ is a basis for $\mathcal{P}_2$, and $p \in \mathcal{P}_2$ is the polynomial with **T** F

$$\mathrm{Rep}_B(p) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

then the set $\{x^2, x, p\}$ is linearly dependent.

(4) There is a unique function $f : \mathbb{R} \to \mathbb{R}$ of the form $f(x) = ax^2 + bx + c$ whose graph passes through the two points $(1, 0), (-1, 0)$. T **F**

(5) If $h : V \to W$ is a linear map and $\dim(V) \geq \dim(W)$, then $h$ must be surjective. T **F**

(6) Let $h : V \to V$ be a linear transformation and $B$ a basis for $V$ such that T **F**

$$\mathrm{Rep}_{B,B}(h) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Then $h$ is an isomorphism.

(7) There exists a surjective linear map $h : \mathbb{R}^3 \to \mathbb{R}^2$ whose null space is 2 dimensional. T **F**

(8) The following matrices are similar. T **F**

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

(9) There is a unique value of $k$ such that the system **T** F

$$\begin{aligned} x + y &= 1 \\ 2x + 2y &= k \end{aligned}$$

has infinitely many solutions.

(10) If **T** F

$$A = \begin{pmatrix} -5/3 & 4/3 \\ -4/3 & 5/3 \end{pmatrix},$$

then $A^{2019} = A$.

**Problem 2** (10 points)**.** In each of the following, a vector space $V$ and a subset $U$ is described. If $U$ is a subspace of $V$, calculate $\dim(U)$ and write its value in the right column. If $U$ is *not* a subspace of $V$, write "N/A" in the right column. You get 1 point for each correct answer. No explanations are required.

| | **Description of $V$ and $U$** | $\dim(U)$ **or N/A** |
|---|---|---|
| (1) | $V = \mathbb{R}^2$ and $$U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x, y \geq 0 \right\}.$$ | **N/A** |
| (2) | $V = \mathbb{R}^3$ and $$U = \left\{ \begin{pmatrix} x \\ x-y \\ y \end{pmatrix} \in \mathbb{R}^3 : x, y \in \mathbb{R} \right\}.$$ | **2** |
| (3) | $V = \mathbb{R}^2$ and $U$ is the null space of the linear map $h : \mathbb{R}^2 \to \mathbb{R}^2$ that first rotates elements $90°$ counterclockwise, and then projects onto the $y$-axis. | **1** |
| (4) | $V = \mathcal{P}_3$ and $U$ is the set of polynomials whose degree is exactly 3. | **N/A** |
| (5) | $V$ is the vector space of all functions $\mathbb{R} \to \mathbb{R}$, and $U$ is the set of all linear combinations of functions $\sin^2(x)$ and $\cos^2(x)$. | **2** |
| (6) | $V = \mathcal{P}_2$ and $U$ is the null space of the function $h : \mathcal{P}_2 \to \mathbb{R}^2$ given by $$p \mapsto \begin{pmatrix} p(0) \\ p(1) \end{pmatrix}.$$ | **1** |
| (7) | $V = \mathcal{M}_{2\times 2}$ and $U$ is the range of the function $h : \mathbb{R}^3 \to \mathcal{M}_{2\times 2}$ given by $$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \mapsto \begin{pmatrix} a+b & a+b \\ a+c & 0 \end{pmatrix}.$$ | **2** |
| (8) | $V = \mathbb{R}^{17}$ and $U$ is the null space of a map $h : \mathbb{R}^{17} \to \mathbb{R}^{34}$ whose rank is 16. | **1** |
| (9) | $V = \mathbb{R}^3$ and $U$ is the range of the function $h : \mathbb{R}^3 \to \mathbb{R}^2$ given by $$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x+1 \\ y+z \end{pmatrix}.$$ | **N/A** |
| (10) | $V = \mathcal{P}_3$ and $U$ is the eigenspace corresponding to the eigenvalue $\lambda = 0$ of the linear transformation $d/dx : \mathcal{P}_3 \to \mathcal{P}_3$. | **1** |

**Problem 3.** Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 3 & 1 \end{pmatrix}$.

(a) (1 point) What is the determinant of $A$?

**Solution.** The matrix is lower triangular, so the determinant is the product of the diagonal entries. So the determinant is equal to 1.

(b) (2 points) Is $A$ invertible? If so, calculate its inverse. If not, explain why not.

**Solution.** The determinant is nonzero, so it must be invertible. To calculate the inverse, we set up the augmented matrix $\begin{pmatrix} A & | & I \end{pmatrix}$ and then row reduce till we get $\begin{pmatrix} I & | & A^{-1} \end{pmatrix}$ where

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 3 & -3 & 1 \end{pmatrix}.$$

**Problem 4** (3 points)**.** Consider the linear map $f : \mathbb{R}^3 \to \mathcal{P}_2$ given by

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \mapsto a + (a+b)x + (a+c)x^2.$$

Is $f$ an isomorphism? If so, prove it. If not, explain why not.

**Solution.** It is an isomorphism. Observe that if $a + (a+b)x + (a+c)x^2 = 0$, then $a = a + b = a + c = 0$, which forces $a = b = c = 0$. Thus $f$ is injective. Since $\dim(\mathbb{R}^3) = \dim(\mathcal{P}_2)$, the rank-nullity theorem implies that $f$ is an isomorphism.

**Problem 5** (3 points)**.** Let $h : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear map given by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} 2x + y + z \\ 3y + z \\ 2z \end{pmatrix}.$$

Is $h$ diagonalizable? If so, find a basis $B$ such that $\mathrm{Rep}_{B,B}(h)$ is diagonal. If not, explain why not.

**Solution.** The matrix of this linear map with respect to the standard bases is

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

This matrix is upper triangular, so its characteristic polynomial is $(2 - x)^2(3 - x)$. In other words, its eigenvalues are 2 with algebraic multiplicity 2 and 3 with algebraic multiplicity 1. The geometric multiplicity of 3 must be equal to 1.

The 2-eigenspace is the null space of

$$A - 2I = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

which means that 2-eigenspace is spanned by $(1, 0, 0)$ and $(0, 1, -1)$. In other words, the geometric multiplicity of 2 is also equal to 2, so the matrix is diagonalizable.

To find the basis $B$, we already have 2 eigenvectors. To find the third, we calculate a basis for the 3-eigenspace. This is the null space of

$$A - 3I = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

which is spanned by $(1, 1, 0)$.

Thus if we take

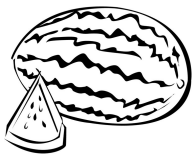
$$B = \langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \rangle$$

then

$$\mathrm{Rep}_{B,B}(h) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

which is diagonal.

**Problem 6** (3 points)**.** On a distant planet, there are red, green, and black tarboojoids. Whenever two differently colored tarboojoids meet, they both change to the third color. If there are initially 70 red, 20 green, and 10 black tarboojoids, can it ever happen that they all turn red? Explain your answer.



**Solution.** We follow the same method as Hefferon's solution to One.I.2.32. Let $a$ be the number of interactions between red and green tarboojoids, $b$ the number between green and black tarboojoids, and $c$ the number between red and black. We are looking for nonnegative integers $a, b, c$ such that

$$\begin{pmatrix} 100 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 70 \\ 20 \\ 10 \end{pmatrix} + a \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} + b \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} + c \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}.$$

We row reduce to solve this system.

$$\left(\begin{array}{ccc|c} -1 & 2 & -1 & 30 \\ -1 & -1 & 2 & -20 \\ 2 & -1 & -1 & -10 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & -1 & 10/3 \\ 0 & 1 & -1 & 50/3 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

In other words, the real solutions are parametrized by the free variable $c$, and we have $a = c + 10/3$ and $b = c + 50/3$. But if $c$ is an integer, then $a$ and $b$ cannot be integers! So there are no integer solutions to this system.

**Problem 7** (Challenge problem!)**.** Let $\mathbb{R}^+$ denote the set of positive real numbers. On one of the problem sets, you showed that $\mathbb{R}^+$ is a vector space where the vector addition operation '$x + y$' is interpreted to mean the product $xy$, and where the scalar multiplication operation '$r \cdot x$' for a scalar $r \in \mathbb{R}$ is interpreted to mean the exponentiation $x^r$.

(a) (1 point) What is the zero element of this vector space? Explain briefly.

**Solution.** The zero element is the element with the property that adding it to another vector does not change the other vector. In this case, the vector addition operation is really multiplication, so the zero vector is the number 1.

(b) (2 points) What is the dimension of this vector space? Explain briefly.

**Solution.** There are various solutions. One is to observe that as $r$ varies, the values of $2^r$ vary over all positive real numbers. Thus the number 2 alone spans $\mathbb{R}^+$, so $\dim(\mathbb{R}^+) = 1$.

Another solution is to consider the map $\log : \mathbb{R}^+ \to \mathbb{R}$. This is an isomorphism. Indeed, we know from pre-calculus that log is bijective, that $\log(xy) = \log(x) + \log(y)$ for all $x, y \in \mathbb{R}^+$, and that $\log(x^r) = r \log(x)$ for $x \in \mathbb{R}^+$ and $r \in \mathbb{R}$. Thus $\dim(\mathbb{R}^+) = \dim(\mathbb{R}) = 1$.

**Honor code.** If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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